Pathogens & disease

How infection spreads

1 Four pathogen groups

Bacteria: cells multiply and may release toxins. **Viruses:** reproduce inside living cells, damaging them.

Fungi and **protists** can cause disease too. Communicable diseases pass between organisms.

2 Break the transmission route

Contact → hygiene/isolation. Contaminated water/food → sanitation, clean preparation and cooking.

Airborne droplets → reduce exposure/isolate infected people. Vectors → prevent bites and breeding.

Vaccination reduces susceptible hosts; match prevention to the pathogen's route.

Viral infections

3 Measles

Virus spread by **inhaling droplets** from coughs/sneezes.

Fever and red rash; complications can be fatal.

Vaccination in childhood reduces illness and spread.

4 HIV → immune damage

Spread by **sexual contact or exchanged body fluids,** including shared needles. Initially flu-like illness.

Attacks immune cells; **AIDS** is severe late-stage damage, with vulnerability to infections/cancers.

Antiretroviral drugs control multiplication. Condoms and avoiding needle sharing reduce spread.

Other pathogens

5 Bacterial infections

Salmonella: contaminated food; bacteria/toxins cause fever, cramps, vomiting and diarrhoea. Hygiene/cooking and poultry vaccination help.

Gonorrhoea: sexual contact; yellow/green discharge and painful urination. Condoms and suitable antibiotics help; resistant strains limit treatment.

6 Malaria

A **protist** causes recurrent fever and can be fatal. The mosquito is the **vector,** not the pathogen.

Life cycle includes mosquitoes. Nets prevent bites; controlling breeding reduces vectors.

Plant virus/fungus examples: **TMV** and **rose black spot;** see plant-health map.